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Chemical kinetic modelling of ammonia/hydrogen/air ignition, premixed flame propagation and NO emission

Rodolfo Cavaliere da Rocha^a, Mário Costa^{a,*}, Xue-Song Bai^b^a IDMEC, Mechanical Engineering Department, Instituto Superior Técnico, Universidade de Lisboa, Lisboa, Portugal^b Division of Fluid Mechanics, Lund University, P.O. Box 118, S-22100 Lund, Sweden

ARTICLE INFO

Keywords:

Ammonia/hydrogen/air flames
Chemical kinetic modelling
Ignition
Flame speed
NO emission

ABSTRACT

This work reports on a study of chemical kinetic modelling of ammonia/hydrogen/air ignition, premixed flame propagation and NO emission. A survey of chemical mechanisms available in the literature was first conducted, and the performance of 10 mechanisms was analysed in terms of the prediction of shock tube ignition delay times, laminar flame speeds and NO_x concentrations for NH₃/air and NH₃/H₂/air flames for a wide range of operating conditions. Then, three of these mechanisms were reduced and their performance compared against the behaviour of the original mechanisms. The results confirm that pure NH₃ flames have high ignition delay times and rather low flame speeds, and that the addition of H₂ to NH₃ flames increases exponentially the flame speed, and significantly the NO_x emission. The currently available chemical kinetic mechanisms predict rather scattered ignition delay times, laminar flame speeds, and NO_x concentrations in NH₃ flames, indicating that improvements in the sub-mechanisms of NH₃ and NH₃/H₂ oxidation are still needed. Sensitivity analysis for NO formation indicates that NO formation in NH₃ flames is mainly produced through the NH₃/O₂ chemical process, and sensitivity analysis for flame speed reveals that the differences among mechanisms are due to the relative importance of the reactions of the NNH and HNO sub-mechanisms. The reduced mechanisms show high fidelity when compared with the original ones, despite some discrepancies at high pressures.

1. Introduction

Ammonia (NH₃) has been proposed as an alternative fuel for a range of applications, including internal combustion engines, gas turbines and space propulsion systems [1,2]. As a flammable substance that does not contain carbon, its combustion does not emit carbon dioxide (CO₂). Production of NH₃ is cost-effective and can be achieved through a number of methods, including from biomass and waste [3]. In addition, NH₃ is considered a safe hydrogen (H₂) carrier due to its low reactivity and can be decomposed onsite into H₂ through catalytic processes [4]. However, NH₃ presents very low flame speeds [5–7], being quite challenging to ignite and burn stably. Moreover, the combustion of NH₃ produces very high amounts of nitrogen oxides (NO_x), especially under lean conditions.

Very recently, Valera-Medina et al. [8] presented a comprehensive review on the use of NH₃ as a viable energy vector for power applications. Indeed, in the last few years, a number of experimental and numerical studies have focused on the use of NH₃ as a fuel, particularly in gas turbines [2,9–14], spark ignition engines [15–19] and compression ignition engines [20]. The studies on combustion engines usually

focused on co-firing NH₃ with kerosene or H₂ [15,17,19]. In these cases, NO_x emissions are usually mitigated using selective catalytic reduction methods, which are able to reduce them to gasoline levels [16,19]. In the case of gas turbines, both pure NH₃ and its mixtures with methane (CH₄) and H₂ have been considered as fuels. In the first category, relevant works include those conducted at Tohoku University, in Japan, which focused on the development of low-NO_x gas turbine combustors through experimental and numerical studies [11,13]. This research group also performed work with NH₃/CH₄ swirl flames, demonstrating that lower NO_x levels are achievable with very low or very high NH₃ fractions in the fuel mixture [12]. A number of experimental and numerical studies on NH₃/CH₄ and NH₃/H₂ flames were conducted at Cardiff University, in United Kingdom, using a high pressure premixed swirl burner [2,9,14]. This research group also focused on chemical kinetics, in particular on the reduction of mechanisms and its coupling with large eddy simulation (LES) for future gas turbine studies [10,21–25].

At a more fundamental level, several studies have been recently performed on the combustion of NH₃. Mathieu and Petersen [7] studied experimentally and numerically the ignition delay times of mixtures of

* Corresponding author at: Mechanical Engineering Department, Instituto Superior Técnico, Av. Rovisco Pais, 1049-001 Lisboa, Portugal.
E-mail address: mcosta@ist.utl.pt (M. Costa).

NH₃/O₂, with 98% and 99% of dilution in argon, in a shock tube at mean pressures of 1.4, 11 and 30 bar and initial temperatures ranging from 1564 to 2489 K. The authors developed a mechanism, based on that of Dagaut et al. [26], able to predict well the measured ignition delay times. For laminar premixed flames, there are a large number of studies available in the literature [5,6,27–44]. Hayakawa et al. [5] examined experimentally the laminar flame speed of mixtures of NH₃ and air, with equivalence ratios ranging from 0.8 to 1.2 at pressures of 0.1, 0.3 and 0.5 MPa and initial temperature of 298 K, in a constant volume cylindrical combustion chamber. The authors concluded that the measured flame speed profiles presented a similar behaviour to those of hydrocarbons flames, but with very low values, with a maximum of ~7 cm/s at 0.1 MPa. The studies of Takizawa et al. [35] and Pfahl et al. [36] showed similar results for spherical flames at atmospheric pressure. Ichikawa et al. [32] evaluated experimentally the stoichiometric flame speeds of NH₃/H₂/air mixtures at pressures of 1, 3 and 5 bar and temperature of 298 K using the same facility used by Hayakawa et al. [5], and concluded that the flame speed increases exponentially as the H₂ content in the mixture increases. Similar conclusions were reported by Lee et al. [28], Li et al. [31] and Kumar and Meyer [33]. Finally, Nozari et al. [34] reported an interesting study of the combustion of mixtures of NH₃/H₂ in a porous media burner.

The chemical kinetic modelling of pure NH₃ and NH₃/H₂ flames has received much attention in recent times. Konnov [45] developed one of the first mechanisms for small hydrocarbons combustion with detailed nitrogen related sub-mechanisms, containing 129 species and 1231 reactions. Mendiara and Glarborg [39] also developed a mechanism for NH₃ chemistry under oxy-CH₄ combustion conditions, containing 97 species and 779 reactions. Tian et al. [40] published an experimental and kinetic modelling study of premixed NH₃/CH₄/O₂/Ar flames, in which they added hydrocarbon/amine reactions to the previous mechanisms, ending with 84 species and 703 reactions. Klippenstein et al. [46] studied the role of NNH in NO formation, applied to studies of NO reduction through NH₃ injection, using a mechanism with 33 species and 207 reactions. Duynslaegher et al. [37] modified the Konnov's mechanism, adding reactions related to the formation of N₂O, but reducing the mechanism to 19 species and 80 reactions. Dagaut et al. [26] studied the oxidation of HCN, which is an important pathway in the formation of NO in the combustion of hydrocarbons and mixtures of hydrocarbons and NH₃, and developed a mechanism containing 69 species and 462 reactions. Mathieu and Petersen [7] reduced and improved Dagaut's mechanism with a novel H₂/O₂ sub-mechanism, ending with 55 species and 278 reactions, whose validation against shock tube ignition data gave excellent results. Song et al. [41] studied the oxidation of NH₃ at high pressures (30 and 100 bar), and temperatures ranging from 450 to 925 K using a mechanism based on that of Klippenstein et al. [46] updated with reactions from other authors, especially with NH₂ + NO_x reactions, resulting in a mechanism with 34 species and 204 reactions. More recently, Okafor et al. [43], Otomo et al. [44] and Glarborg et al. [47] focused their studies on the usage of NH₃ as fuel. Okafor et al. [43] developed a mechanism, with 59 species and 356 reactions, for the combustion of mixtures of NH₃ and CH₄, specifically for laminar burning velocity studies, combining the GRI Mech 3.0 mechanism [48] and that of Tian et al. [40]. Otomo et al. [44] developed a mechanism, with 59 species and 356 reactions, for the combustion of NH₃/air and NH₃/H₂/air based on that of Song et al. [41], with improved reactions for NH₂, HNO and N₂H₂. Finally, Glarborg et al. [47] developed a detailed mechanism for nitrogen chemistry in combustion, with 151 species and 1397 reactions, updating the nitrogen compounds' thermochemistry, and being able to describe the prompt-, thermal- and fuel-NO formation and the NO formation through the NNH or N₂O mechanisms as well as the NO removal processes.

Proper and cost-effective large eddy simulation (LES) and direct numerical simulation (DNS) of gas turbines and spark ignition engines require kinetic mechanisms with reduced number of species and

reactions [49]. The present work aims i) to examine the performance of the various existing mechanisms in predicting critical features such as ignition delay times, laminar flame speeds and NO formation for NH₃/H₂/air premixed flames, and ii) to reduce selected mechanisms through a newly developed algorithm, for use in computational fluid dynamics codes.

2. Computational details

The mechanisms considered in this study for performance evaluation were those developed by Konnov [45], Mendiara and Glarborg [39], Klippenstein et al. [46], Duynslaegher et al. [37], Dagaut et al. [26], Mathieu and Petersen [7], Song et al. [41], Okafor et al. [43], Otomo et al. [44] and Glarborg et al. [47]. Subsequently, the mechanisms of Mathieu and Petersen [7], Okafor et al. [43] and Otomo et al. [44] were selected for mechanism reduction.

All simulations were made in Cantera [50]. Cantera is an open-source package of object-oriented tools for the solution of problems involving chemical kinetics, thermodynamics and transport problems. It has automated functions for the resolution of laminar flames, reactors, manipulation of chemical kinetics and other combustion-related issues. The libraries of Cantera are available in C/C++, Fortran 90, Matlab and Python languages. In this work, the Python language was used, with pyMARS [51] utilities used for mechanisms writing.

2.1. Ignition delay times, flame speeds and NO formation

The shock tube simulation performed in this study is based on an idealized constant volume, zero-dimensional reactor model available in Cantera [50]. A mixture of reactants at a given pressure and temperature in a vessel is submitted to a sudden increase in pressure (a shock), ignites, and the ignition delay time is measured. The experimental data used here for comparisons were taken from Mathieu and Petersen [7], who performed measurements for NH₃/O₂ mixtures in a shock tube, with 98–99% dilution in argon at pressures of 1.4, 11 and 30 bar and temperatures ranging from 1564 to 2489 K.

Laminar premixed flames were simulated using the adiabatic, freely propagating model available in Cantera [50]. The standard mesh refining criteria were selected and meshes were automatically created by the software. Flame speed and NO formation were computed for NH₃/H₂/air flames, with the NH₃/H₂ volume ratio ranging from 0 to 1, under one-dimensional freely propagating flame conditions, with the premixed reactants injected at pressures of 1 and 5 bar, temperature of 298 K and fuel/air equivalence ratios ranging from 0.5 to 1.5.

2.2. Sensitivity analysis

A sensitivity analysis for the mechanisms was performed for one-dimensional premixed laminar flame speed and for NO production in a model reactor. The Cantera built-in sensitivity tools were used to quantify the contribution of each reaction in the acceleration and deceleration of the global reaction (flame speed), and in the formation of NO. For the first, the formulation is:

$$\text{Sensitivity} = \frac{k_i}{s_f} \times \frac{\partial s_f}{\partial k_i} \quad (1)$$

where s_f is the flame speed and k_i is the reaction rate constant of the i -th reaction. For the second, the formulation is:

$$S_{ki} = \frac{k_i}{y_i} \times \frac{\partial y_i}{\partial k_i} \quad (2)$$

where y_i is the mole fraction of the species under evaluation.

For the flame speed sensitivity analysis, one-dimensional premixed laminar flames, with fuel mixtures containing 0%, 50% and 100% of H₂, at 1 bar and initial temperature of 298 K under stoichiometric conditions were evaluated. For the NO production sensitivity analysis,

an ideal zero-dimensional constant pressure reactor operating with fuel mixtures containing 0%, 20% and 80% of H_2 at a temperature of 1500 K, pressures of 1 and 5 bar and equivalence ratios of 0.8, 1 and 1.2 was examined.

2.3. Mechanism reduction

The reduced mechanisms must follow specific criteria [49], namely, (1) to precisely model the properties of the original system deemed relevant to the problem under study, (2) to be as stable as the original one, (3) to be simpler and demand lower computational power than the original one, (4) to keep non-linear behaviours from the original mechanism, and (5) to correlate to the physical phenomena, correctly describing the problem and generating insights on it.

A mechanism containing n chemical species, with mass fractions y_i ($i = 1, \dots, n$), and N reactions, with the reaction rate of the r -th elementary reaction being $F_r(y)$ ($r = 1, \dots, N$), can be considered a system for a constant volume reactor given by:

$$\dot{y} = g(y) = \sum_{r=1}^N S_r F_r(y) \quad (3)$$

where $S_r \in \mathbb{R}^n$ is the vector of the species coefficients for the elementary reaction rates F_r . For example, for the reaction $O_2 + H \leftrightarrow OH + O$, $F_1 = k_{1,f}[O_2][H] - k_{1,b}[OH][O]$, where $k_{1,f}$ and $k_{1,b}$ are the rate constants of the forward and reverse reactions, respectively.

The S_r vectors can be grouped in a matrix, S , while the non-linear reaction rates F_r can be grouped in a vector, F , the kinetic mechanism for $S \in \mathbb{R}^{n \times N}$ and $F \in \mathbb{R}^N$. In this manner, the original system can be written as:

$$\dot{y} = SF(y) \quad (4)$$

This equation system can be reduced multiplying it by a matrix D :

$$\dot{z} = SDF(y) \quad (5)$$

where $D \in \mathbb{R}^{N \times N}$ is a diagonal matrix, whose d_i elements have the value of 1 if the i -th reaction is selected or 0 if the i -th reaction is not selected.

The present algorithm (see Fig. S1 in the Supplementary Document) is based on the mechanism reduction method of Petzhold and Zhu [49] and a built-in code from Cantera [50]. A zero-dimensional shock tube ignition simulation is run for a certain pressure and temperature condition, and the maximum absolute values of the net rates of progress, i.e., the rate of the species formation, dy_i/dt , for each reaction in any time step is recorded as a sensitivity coefficient of that reaction. The mechanism, F , is then re-ordered from the highest to the lowest maximum dy_i/dt , becoming F_{ν} . Next, ignition simulations are repeated iteratively, selecting sub-mechanisms in F_{ν} , using reactions from 0 to k , with $k = 1, \dots, N$, l being the minimum number of reactions that keep all selected species (in this case Ar, NH_3 , O_2 , N_2 , NO, H_2O , OH, H and H_2) and N the number of reactions in the full mechanism. The optimized reduced mechanism is the one that originates the closest temperature profile and ignition delay time to those of the original mechanism. The fitting parameters are given as follows:

$$\sqrt{\frac{\sum_{t=1}^{t_f} \frac{(T_{orig,t} - T_{red,t})^2}{T_{orig,t}^2}}{t_f}} \leq 0.001 \quad (6)$$

where t is the time, t_f is the time at the end of the simulation, and $T_{orig,t}$ and $T_{red,t}$ are the instantaneous temperatures in the original and in the reduced ignition profile simulation, respectively,

$$\frac{|t_{ign,orig} - t_{ign,red}|}{t_{ign,orig}} \leq 0.001 \quad (7)$$

where $t_{ign,orig}$ and $t_{ign,red}$ are the original and the reduced ignition delay time, respectively.

In this work, the conditions used for the shock tube ignition were a

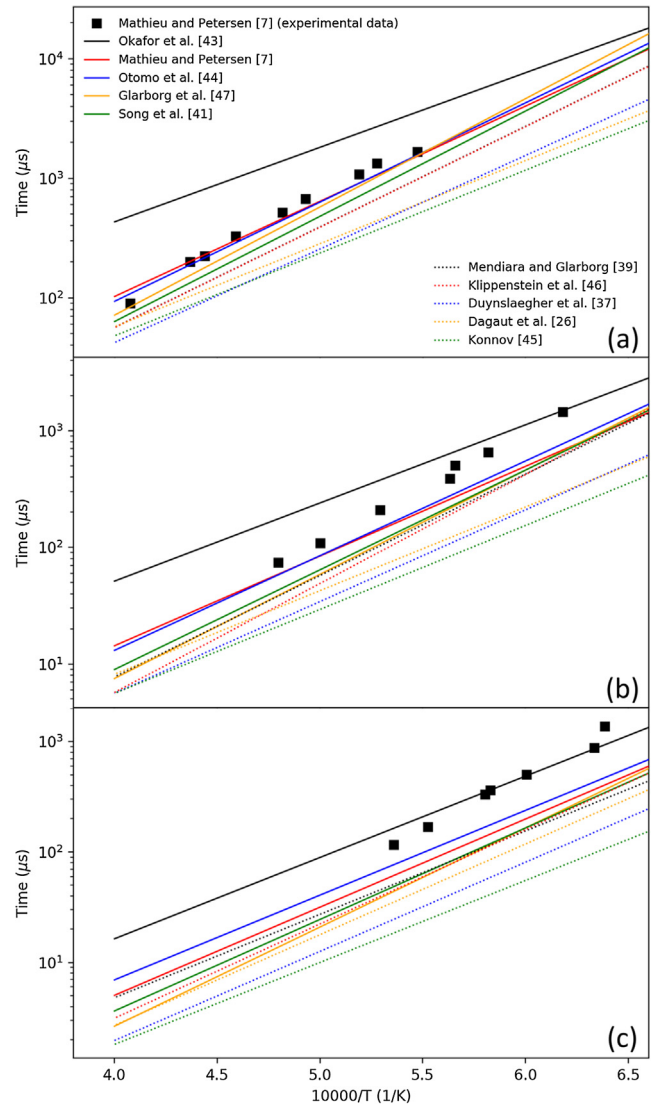


Fig. 1. Shock tube ignition delay times for stoichiometric NH_3/O_2 mixtures with 98% of dilution in Ar. (a) 1.4 bar, (b) 11 bar, (c) 30 bar.

pressure of 1.4 bar, a temperature of 1927 K, a NH_3/O_2 equivalence ratio of 0.5 and a dilution in argon of 99%.

3. Results and discussion

3.1. Ignition delay time

Fig. 1 shows predictions of ignition delay times, obtained with 10 mechanisms, along with the experimental data obtained by Mathieu and Petersen [7], for stoichiometric NH_3/O_2 mixtures with 98% of dilution in Ar at 1.4, 11 and 30 bar. Figs. S2–S4 in the Supplementary Document show data for other conditions. All figures show that the various mechanisms yield quite different results, with most of them underpredicting the ignition delay times as compared with the experimental data. Overall, the mechanisms of Mathieu and Petersen [7] and Otomo et al. [44] are those that predict better the experimental data. The mechanism of Glarborg et al. [47] performs reasonably well for some conditions, but it underpredicts the ignition delay times for most cases, while the mechanism of Okafor et al. [43] overpredicts the ignition delay times for all conditions.

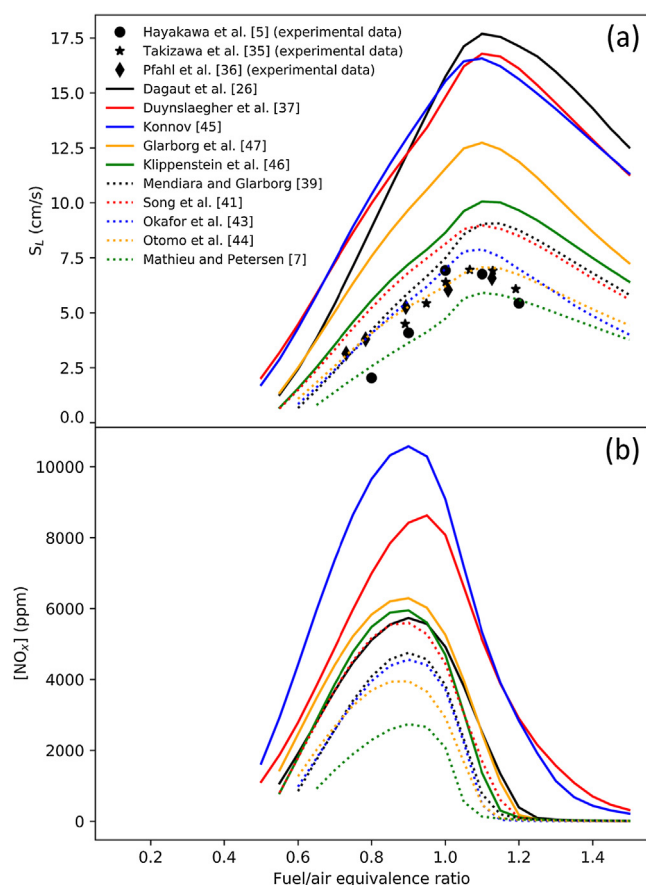


Fig. 2. (a) Laminar flame speeds and (b) NO_x concentrations for NH_3 /air mixtures as a function of the equivalence ratio at 1 bar and 298 K.

3.2. Flame speed and NO formation

Fig. 2 shows predictions of laminar flame speeds and NO_x concentrations, obtained with 10 mechanisms, for NH_3 /air mixtures as a function of the equivalence ratio at 1 bar and 298 K. Fig. S5 in the Supplementary Document shows similar predictions for 5 bar. The figures also include experimental data for flame speed obtained by Hayakawa et al. [5], Takizawa et al. [35] and Pfahl et al. [36]. At 1 bar, all mechanisms predict flame speeds higher than those measured experimentally, except those of Mathieu and Petersen [7], Okafor et al. [43] and Otomo et al. [44]. The mechanism of Mendiara and Glarborg [39] only performs well from lean to stoichiometric conditions, overpredicting the flame speeds under rich conditions. The mechanisms predict quite different NO_x concentrations, with a minimum peak value around 2800 ppm for the mechanism of Mathieu and Petersen [7] and a maximum peak of 10500 ppm for that of Konnov [45]. At 5 bar (Fig. S5), similar results are obtained, and only the mechanisms of Mathieu and Petersen [7], Okafor et al. [43] and Otomo et al. [44] predict flame speeds close to the experimental data. It must be noted that no experimental NO_x data for the studied conditions were found in the literature.

Fig. 3 shows predictions of laminar flame speeds and NO_x concentrations, obtained with 10 mechanisms, for stoichiometric NH_3/H_2 /air mixtures as a function of the H_2 mole fraction in the fuel mixture at 1 bar and 298 K. Fig. S6 in the Supplementary Document shows similar predictions for 5 bar. The figures also include experimental data for flame speed obtained by Ichikawa et al. [32], Lee et al. [28], Li et al. [38] and Kumar and Meyer [33]. For stoichiometric NH_3/H_2 /air flames the mechanisms of Glarborg et al. [47], Konnov [45] and Dagaut et al. [26] overpredict the flame speeds for all conditions. The mechanism of

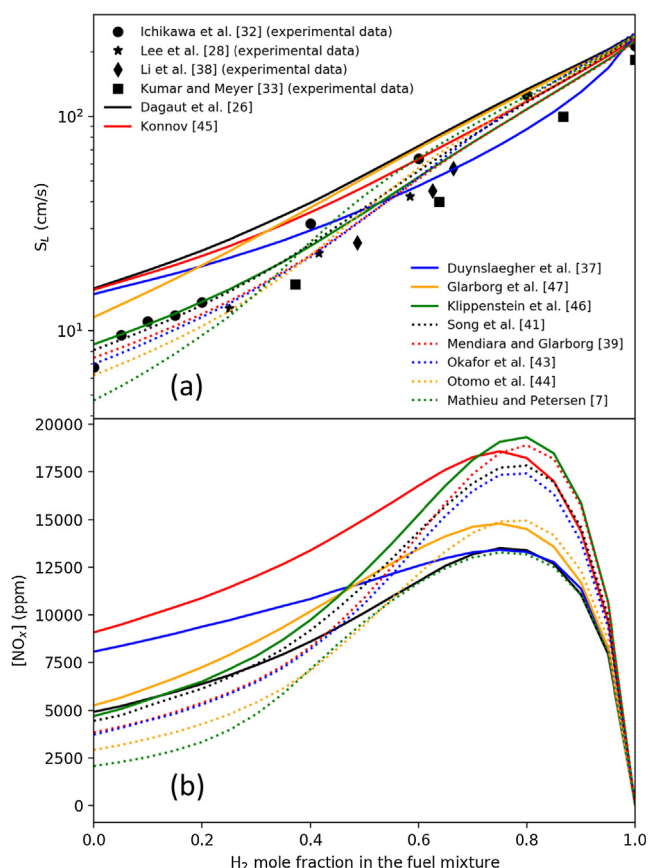


Fig. 3. (a) Laminar flame speeds and (b) NO_x concentrations for stoichiometric NH_3/H_2 /air mixtures as a function of the H_2 mole fraction in the fuel mixture at 1 bar and 298 K.

Duynslaegher et al. [37] overpredicts the flame speeds for low H_2 content in the fuel mixture, underpredicting these values for H_2 contents above 40%. The mechanisms of Klippenstein et al. [46] and Song et al. [41] predict well the behaviour of the mixtures with H_2 contents from 5% to 100%. The mechanisms of Mendiara and Glarborg [39] and Okafor et al. [43] predict higher flame speeds than the measured ones for low H_2 contents in the fuel mixture, predicting values closer to the experimental data for H_2 contents in the fuel mixture higher than ~60%. The mechanism of Mathieu and Petersen [7] shows the poorest agreement with the experiments for mixtures with low H_2 content. The mechanism of Otomo et al. [44] predicts flame speeds in between those predicted by the mechanisms of Mathieu and Petersen [7] and Okafor et al. [43]. It should be pointed out that the laminar flame speed of stoichiometric NH_3/H_2 /air flames increases approximately exponentially with the mole fraction of H_2 in the fuel mixture. Thus, it seems an appropriate approach to improve the flame speed of NH_3 flames through the addition of H_2 to the fuel mixture. It should be noted that for 5 bar the mechanism of Konnov [45] presented poor convergence in Cantera.

Figs. 3 and S6, in the Supplementary Document, reveal an increase in NO_x production with the content of H_2 in the fuel mixture up to 80%, probably owing to the presence of OH and O radicals at high temperatures. For H_2 contents in the fuel mixture higher than 80% the NO_x production decreases, owing to the decreasing content of NH_3 in the blend. Similar trends were obtained for higher pressures.

3.3. Sensitivity analysis

Figs. 4 and 5 show the flame speed sensitivity to the 10 most important reactions in all mechanisms examined for NH_3 /air flames. It is

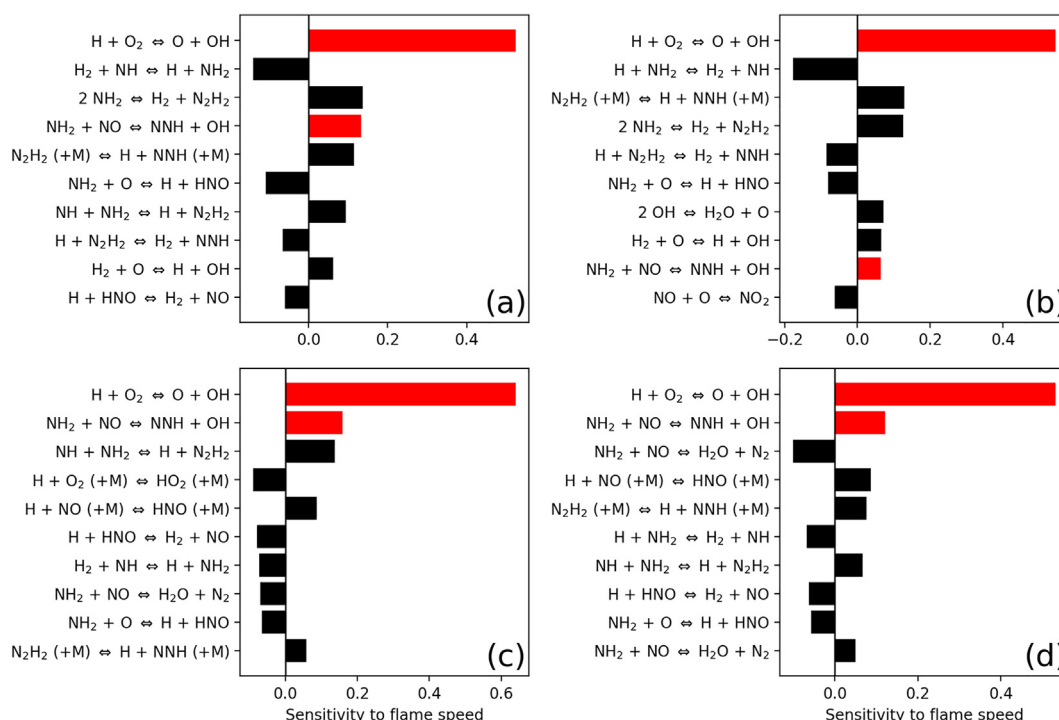


Fig. 4. Flame speed sensitivity for NH₃/air flames at 1 bar, 298 K and stoichiometric conditions for the mechanisms of (a) Konnov [45], (b) Duynslaegher et al. [37], (c) Glarborg et al. [47], (d) Klippenstein et al. [46]. Critical reactions are marked in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

seen that the most important reaction in all mechanisms is $\text{H} + \text{O}_2 \rightleftharpoons \text{O} + \text{OH}$, with a sensitivity value around 0.5. The mechanisms of Mendiara and Glarborg [39], Okafor et al. [43], Otomo et al. [44] and Mathieu and Petersen [7], which predict better the experimental data, show the reaction $\text{H} + \text{NO} + \text{M} \rightleftharpoons \text{HNO} + \text{M}$ as the second most important in the promotion of flame speed in NH₃/air mixtures, with sensitivity values between 0.1 and 0.2. The mechanisms that overpredict the flame speeds have this reaction as less important; this is the case of the mechanisms of Konnov [45] and Dagaut et al. [26], which do not list this reaction among the ten most important, and of the mechanism of Glarborg et al. [47], which provides a sensitivity value of less than 0.1. The mechanisms of Konnov [45], Klippenstein et al. [46] and Glarborg et al. [47] show the reaction $\text{NH}_2 + \text{NO} \rightleftharpoons \text{NNH} + \text{OH}$ as one of the most important, instead, with sensitivity values higher than 0.1.

For mixtures of NH₃/H₂/air (Figs. S7 and S8 in the Supplementary Document), the relative importance of the nitrogen sub-mechanisms reduces gradually, while the importance of the H₂/O₂ reactions augments. Moreover, the reactions between nitrogen containing radicals and OH, such as $\text{NH}_2 + \text{NO} \rightleftharpoons \text{NNH} + \text{OH}$, become more important for all mechanisms.

Figs. 6 and 7 show the sensitivity of NO emission to the 10 most important reactions in NO production or consumption for NH₃/air flames in the mechanisms of Mathieu and Petersen [7] and Glarborg et al. [47]. It is observed that the most important reactions in the mechanism of Mathieu and Petersen [7] in the formation of NO are the same for all conditions ($\text{NH}_2 + \text{NO} \rightleftharpoons \text{NNH} + \text{OH}$ and $\text{H} + \text{O}_2 \rightleftharpoons \text{O} + \text{OH}$), while for the reduction of NO the most important reactions are $\text{NH}_2 + \text{NO} \rightleftharpoons \text{H}_2\text{O} + \text{N}_2$ and $\text{H}_2\text{NO} + \text{NH}_2 \rightleftharpoons \text{NH}_3 + \text{HNO}$. Under lean and stoichiometric conditions, the formation of NO increases with the amount of O, O₂ and OH radicals, which combine with those from the NH₃ decomposition. For the same conditions, the mechanism of Glarborg et al. [47] show similar results, but with much higher sensitivity values for each reaction, and with the reaction $\text{NH}_2 + \text{O}_2 \rightleftharpoons \text{H}_2\text{NO} + \text{O}$ being ranked

as the most important for the production of NO; for lean flames, the mechanism of Glarborg et al. [47] also considers the reaction $2\text{NH}_2 \rightleftharpoons \text{NH}_3 + \text{NH}$ important for NO generation. For rich flames, in the mechanism of Mathieu and Petersen [7], the reactions that present the highest sensitivity values are the same as those in the lean and stoichiometric cases. They contribute to the formation of NO around the ignition time and, later, consume the produced NO. The NO consumption occurs presumably through the reaction with NH₂ radicals, formed by the pyrolysis of NH₃. For this condition, the mechanism of Glarborg et al. [47], however, does not show the same behaviour, presenting lower sensitivity values for the most sensitive reactions. The thermal dissociation of N₂ seems not to be as significant as the other reactions, indicating that the NO formation in NH₃ flames is essentially due to the NH₃ decomposition. At 5 bar, the mechanism of Mathieu and Petersen [7] behaves in a similar way as at lower pressures but producing less NO than at lower pressures. The mechanism of Glarborg et al. [47], however, also ranks the reaction $\text{HNOH} + \text{NH}_2 \rightleftharpoons \text{H}_2\text{NN} + \text{H}_2\text{O}$ as one of the most sensitive at high pressure. This might be due to the capture of O radicals in form of H₂O, avoiding its reaction with nitrogen radicals and thus the formation of NO.

In the case of NH₃/H₂/air flames (Figs. S9 and S10 in the Supplementary Document), the behaviour of the NO sensitivity seems to depend mostly on the fuel composition for both mechanisms, while the time interval in which most NO is produced reduces significantly as the H₂ content in the fuel mixture increases. For 20% of H₂ in the fuel mixture, in the Mathieu and Petersen [7] and Glarborg et al. [47] mechanisms, the reaction $\text{H} + \text{O}_2 \rightleftharpoons \text{O} + \text{OH}$ is much more important for the NO formation than any other, indicating that the NO generation is mainly driven by the abundance of O and OH radicals, which interact with radicals from the NH₃ decomposition, such as NH₂ produced through the reaction $\text{H} + \text{NH}_3 \rightleftharpoons \text{H}_2 + \text{NH}_2$. In this case, the burning process in both mechanisms indicates two stages of NO production: in the first stage, part of the NH₃ and most of the H₂ are decomposed and, in the second stage, the remaining NH₃ reacts with

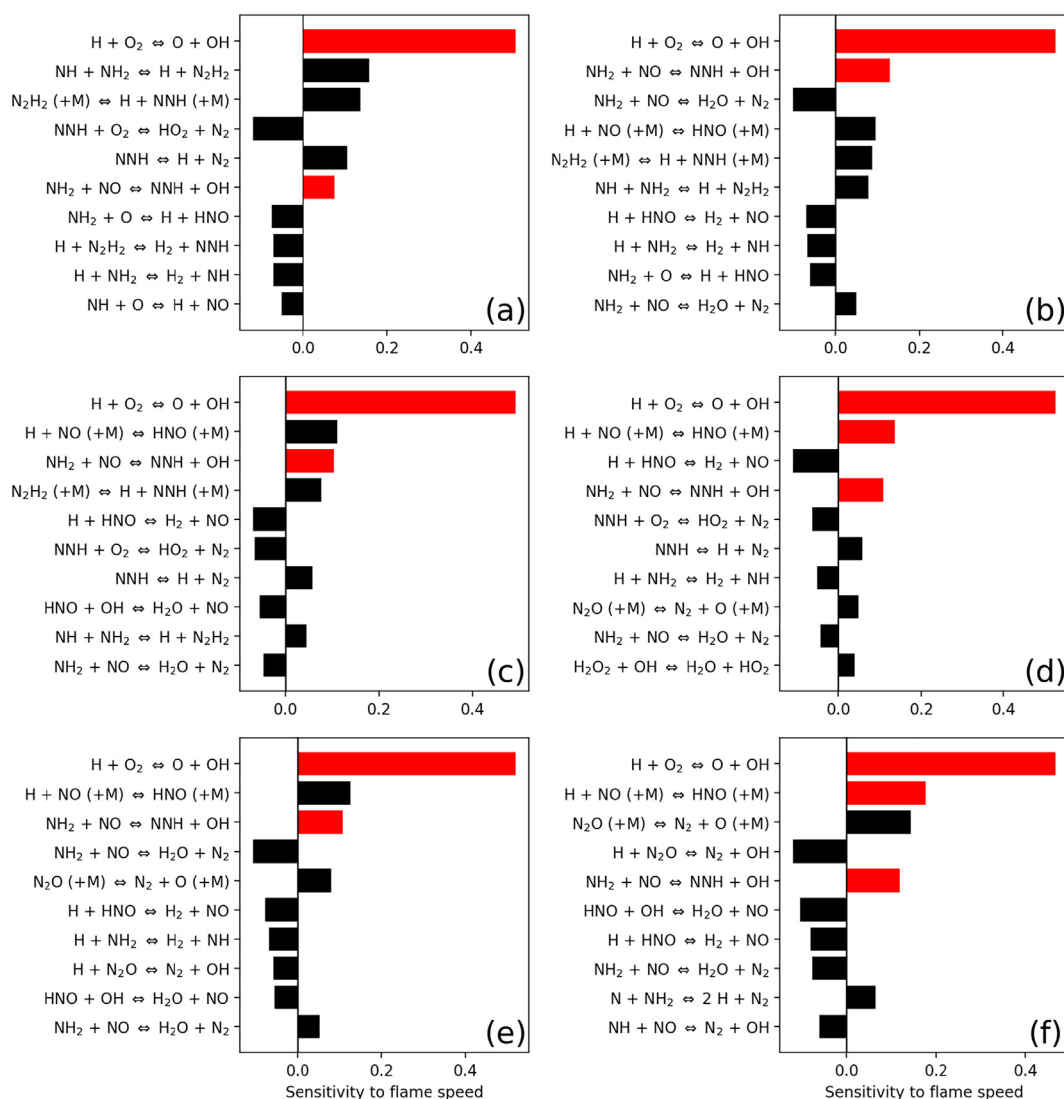


Fig. 5. Flame speed sensitivity for NH_3/air flames at 1 bar, 298 K and stoichiometric conditions for the mechanisms of (a) Dagaut et al. [26], (b) Song et al. [41], (c) Mendiara and Glarborg [39], (d) Okafor et al. [43], (e) Otomo et al. [44], (f) Mathieu and Petersen [7]. Critical reactions are marked in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

radicals produced from the previous H_2 decomposition. It is noticeable that the second stage is much more important in the mechanism of Glarborg et al. [47], with a sensitivity value more than 3 times higher than that of the first stage. For 50% of H_2 in the fuel mixture, the same two-stage process occurs; however, reactions with nitrogen containing radicals from the NH_3 decomposition become less important, and the reaction $\text{N}_2 + \text{O} \rightleftharpoons \text{N} + \text{NO}$, possibly related to thermal NO formation, appears in the direction of the consumption of NO. In this case, both mechanisms present similar overall results, but in the mechanism of Glarborg et al. [47] the reaction $\text{HNO} + \text{O}_2 \rightleftharpoons \text{HO}_2 + \text{NO}$ is shown to be the important one for the production of NO. For 80% of H_2 in the fuel mixture, the whole process occurs in a single stage, and smaller nitrogen containing radicals (N and NH) seem to interact with oxygen, generating NO as well as O and OH radicals, which are related to NO formation in other reactions. The NO consumption is related to the reactions $\text{NH}_3 + \text{OH} \rightleftharpoons \text{H}_2\text{O} + \text{NH}_2$ and $\text{H}_2 + \text{O} \rightleftharpoons \text{N} + \text{NO}$, with both mechanisms presenting generally similar results. The mechanism of Glarborg et al. [47], however, presents again the reaction $\text{HNO} + \text{O}_2 \rightleftharpoons \text{HO}_2 + \text{NO}$ as an important one for the production of NO.

3.4. Mechanism reduction

The mechanisms of Mathieu and Petersen [7], Okafor et al. [43] and Otomo et al. [44] were selected for mechanism reduction, considering their performance in predicting the ignition delay times and laminar flame speeds of NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ flames. As a first step, the mechanisms had all carbon-related species and reactions removed, ending up with 34, 32 and 25 chemical species, and 159, 213 and 100 reactions, respectively. By applying the developed algorithm (cf. Fig. S1 in the Supplementary Document), the mechanisms of Mathieu and Petersen [7], Okafor et al. [43] and Otomo et al. [44] were reduced to mechanisms containing 22, 24 and 21 chemical species and 66, 72 and 51 reactions, respectively. Fig. 8 shows the ignition delay time as a function of the number of reactions for the three mechanisms, and Table 1 summarizes the chemical species and reactions in the original and reduced mechanisms.

Fig. 9 shows predictions of ignition delay times, obtained with the original and reduced mechanisms, along with the experimental data obtained by Mathieu and Petersen [7], for stoichiometric NH_3/O_2 mixtures with 98% of dilution in Ar at 1.4, 11 and 30 bar. Figs. S11 and S13 in the Supplementary Document show data for other conditions. It is seen that the reduced mechanisms yield similar results to those of the

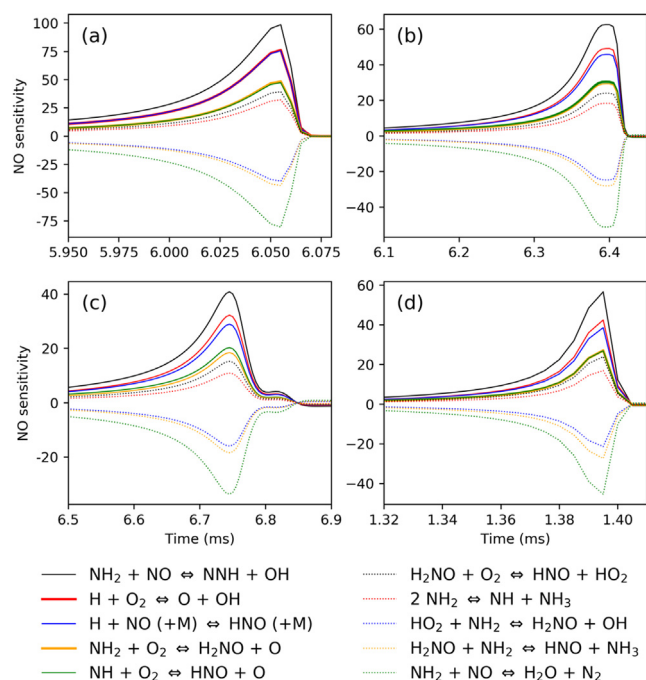


Fig. 6. NO sensitivity for NH_3/air flames in the mechanism of Mathieu and Petersen [7]. (a) 1 bar for an equivalence ratio of 0.8, (b) 1 bar for a stoichiometric mixture, (c) 1 bar for an equivalence ratio of 1.2, (d) 5 bar for a stoichiometric mixture.

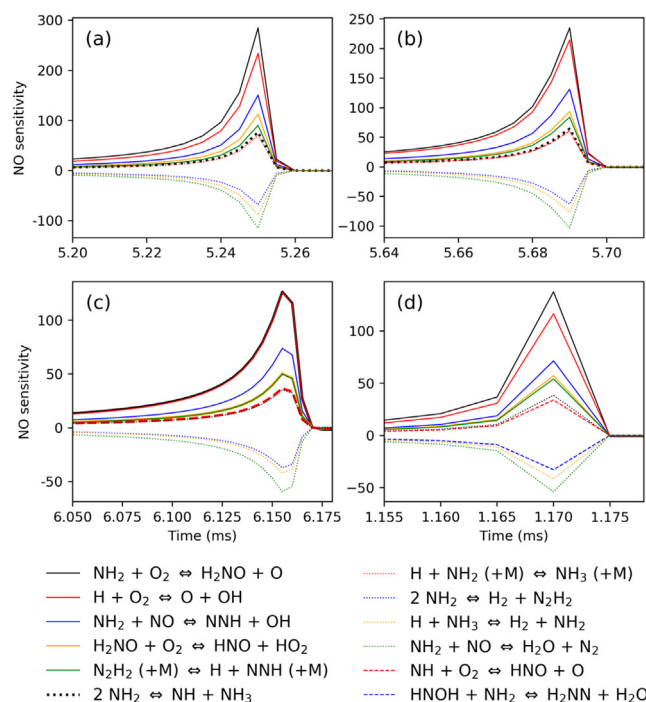


Fig. 7. NO sensitivity for NH_3/air flames in the mechanism of Glarborg et al. [47]. (a) 1 bar for an equivalence ratio of 0.8, (b) 1 bar for a stoichiometric mixture, (c) 1 bar for an equivalence ratio of 1.2, (d) 5 bar for a stoichiometric mixture.

original ones in most cases. The Ok-R51 mechanism results are very close to those of the original one in all cases, with a maximum difference of around 1500 μs for an equivalence ratio of 2 at 1.4 bar and 1500 K. The MP-R66 mechanism also yields similar results to those of the original mechanism. The Ot-R72 mechanism, however, shows larger differences, up to 19000 μs for stoichiometric conditions.

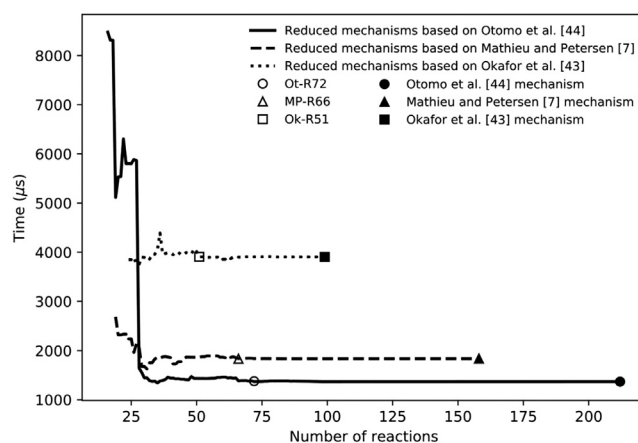


Fig. 8. Ignition delay time as a function of the number of reactions for the mechanism of Mathieu and Petersen [7], Okafor et al. [43] and Otomo et al. [44].

Fig. 10 shows predictions of laminar flame speeds and NO_x concentrations, obtained with the original and reduced mechanisms, for NH_3/air mixtures as a function of the equivalence ratio at 1 bar and 298 K. Fig. S14 in the Supplementary Document shows similar predictions for 5 bar. The figures also include experimental data for flame speed obtained by Hayakawa et al. [5], Takizawa et al. [35] and Pfahl et al. [36]. The results obtained with the reduced mechanisms show good agreement with the original mechanisms in all cases. In terms of flame speed, the MP-R66 and Ok-R51 mechanisms show excellent agreement, with maximum differences of ~ 0.1 cm/s. In the case of the Ot-R72 mechanism, the maximum overall difference in relation to the original mechanism is 1.1 cm/s at an equivalence ratio of 1.45, which is not significant since the mechanisms tend to perform worse for rich flames. In terms of NO_x concentrations, there is also good agreement between the mechanisms of Okafor et al. [43] and Mathieu and Petersen [7] and the respective reduced mechanisms. For the former, the maximum difference is ~ 200 ppm at stoichiometry, and for the latter a maximum difference of ~ 40 ppm is observed at 5 bar for an equivalence ratio of 0.9. In the case of the mechanism of Otomo et al. [44], however, a maximum difference of ~ 660 ppm is detected for an equivalence ratio of 0.8.

Fig. 11 shows predictions of laminar flame speeds and NO_x concentrations, obtained with the original and reduced mechanisms, for stoichiometric $\text{NH}_3/\text{H}_2/\text{air}$ mixtures as a function of the H_2 mole fraction in the fuel mixture at 1 bar and 298 K. Fig. S15 in the Supplementary Document shows similar predictions for 5 bar. The figures also include experimental data for flame speed obtained by Ichikawa et al. [32], Lee et al. [28], Li et al. [38] and Kumar and Meyer [33]. Again, a good agreement between the predicted data and the original and reduced mechanisms is observed. For flame speeds, at 1 bar, for H_2 contents in the fuel mixture higher than 30% the mechanisms of Okafor et al. [43] and Ok-R51 diverge, with the reduced mechanism giving results that exceed those of the original mechanism by 16% for a H_2 content of 50% in the fuel mixture. For the mechanisms of Mathieu and Petersen [7] and Otomo et al. [44] and respective reduced mechanisms, the differences are below 8%. For higher pressures, larger differences are observed for the Ot-R72 and Ok-R51 mechanisms, while the MP-R66 mechanism presents smaller differences (less than 2%) from the original mechanism. In regard to the NO_x concentrations, the reduced mechanisms show similar results to the respective original mechanisms, with the largest differences observed for the Ok-R51 mechanism, regardless of the pressure.

Table 1
Chemical species and reactions in original and reduced mechanisms.

Reference	Original species	Original reactions	Non-C species	Non-C reactions	Species in reduced mechanism	Reactions in reduced mechanism
Mathieu and Petersen [7]	55	278	34	159	22	66
Okafor et al. [43]	59	356	25	100	21	51
Otomo et al. [44]	32	213	32	213	24	72

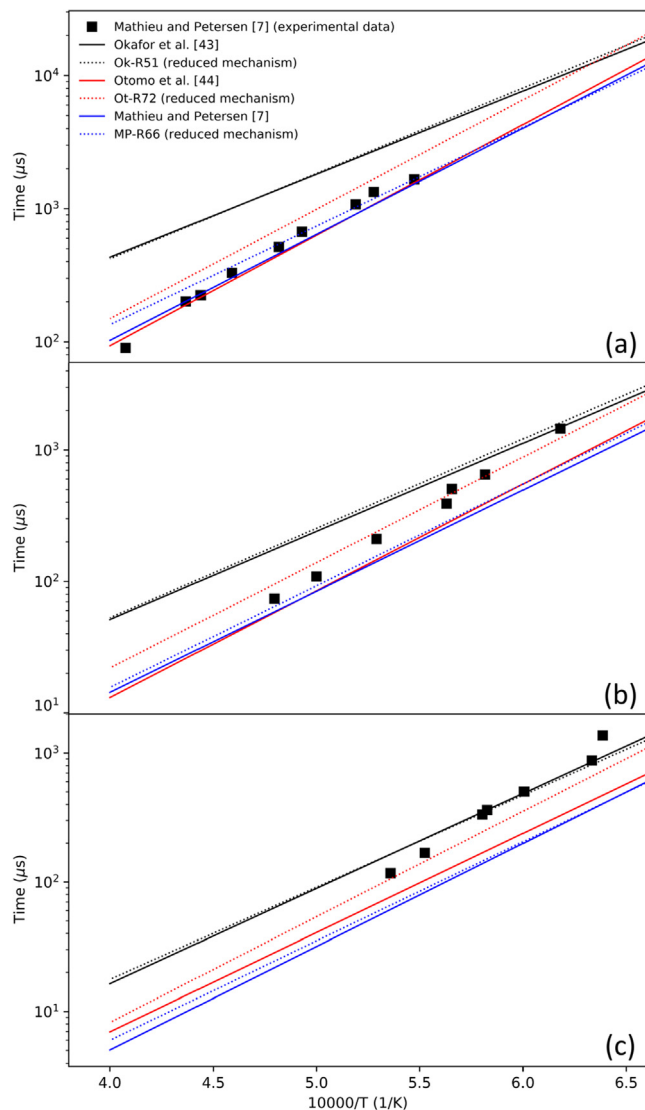


Fig. 9. Shock tube ignition delay times for stoichiometric NH_3/O_2 mixtures with 98% dilution in Ar. (a) 1.4 bar, (b) 11 bar, (c) 30 bar.

4. Conclusions

Ten chemical kinetic mechanisms available in the literature were used to calculate ignition delay times, laminar flame speeds and NO_x production in NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ flames for a wide range of operating conditions. Following comparisons against experimental data on ignition delay times and laminar flame speeds available in the literature, three of these mechanisms were reduced using an algorithm that allowed establishing new mechanisms that represent well the behaviour of the original mechanisms. The main conclusions of this study are as follows.

1) Pure NH_3 flames have high ignition delay times and rather low flame speeds, and the addition of H_2 to NH_3 flames is a good way to

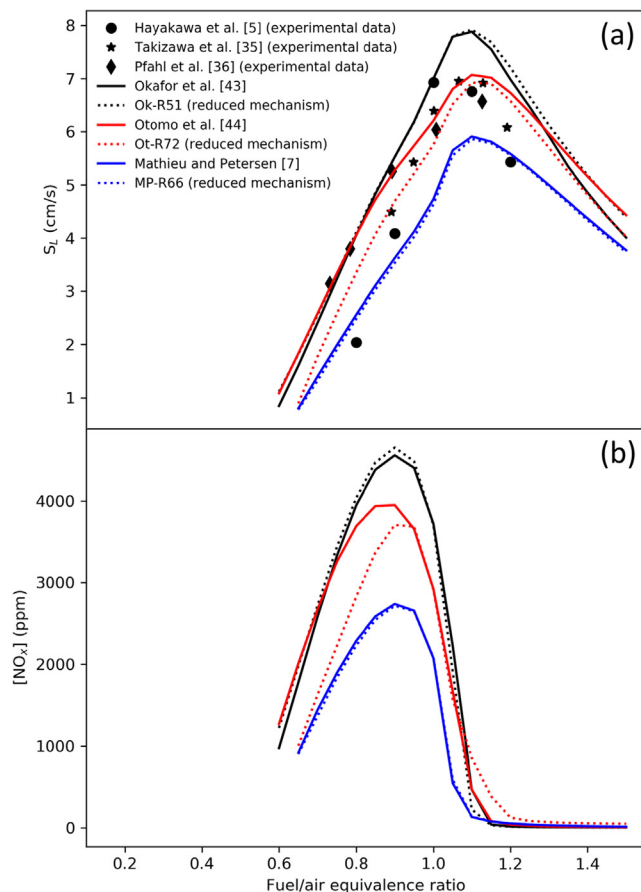


Fig. 10. (a) Laminar flame speeds and (b) NO_x concentrations for NH_3/air mixtures as a function of the equivalence ratio at 1 bar and 298 K.

improve its combustion behaviour.

- 2) NO_x emissions are significantly higher in NH_3/H_2 flames than in pure NH_3 flames, probably due to the generation of OH and O radicals.
- 3) Currently available chemical kinetic mechanisms predict rather scattered ignition delay times, laminar flame speeds, and NO_x concentrations in NH_3 flames, indicating that improvements in sub-mechanisms of NH_3 and NH_3/H_2 oxidation are still needed.
- 4) Sensitivity analysis for NO formation indicates that NO formation in NH_3 flames is mainly produced through the NH_3/O_2 chemical process; excess of NH_3 in flames drive reactions to the consumption of NO after ignition, being a possible route to NO mitigation.
- 5) Sensitivity analysis for flame speed reveals that differences among mechanisms are due to the relative importance of reactions of the NNH and HNO sub-mechanisms.
- 6) The mechanisms available in the literature can still be further reduced with high fidelity when compared with the original ones. However, slightly different behaviours were observed for ignition delay times, flame speeds and NO_x production at high pressures.

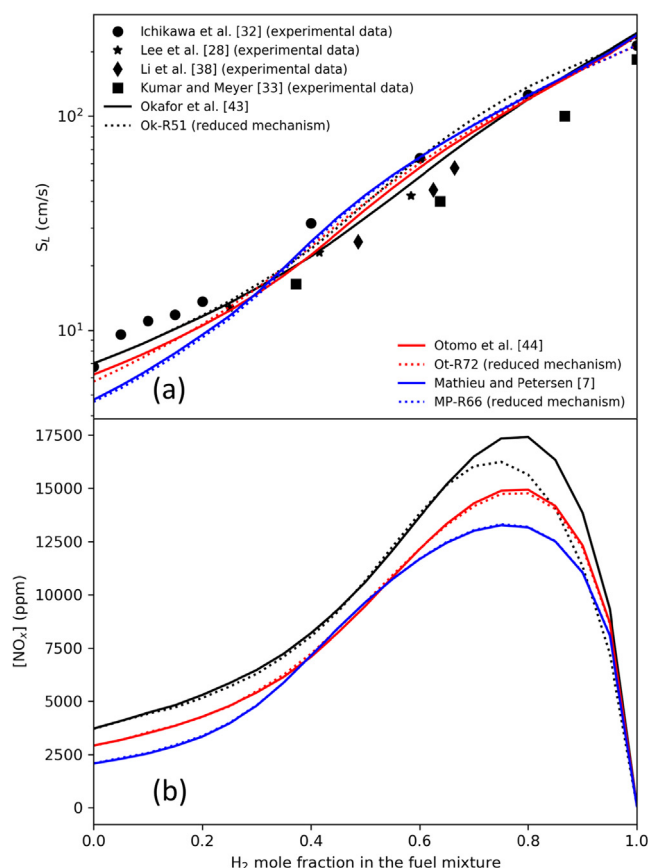


Fig. 11. (a) Laminar flame speeds and (b) NO_x concentrations for stoichiometric NH₃/H₂/air mixtures as a function of the H₂ mole fraction in the fuel mixture at 1 bar and 298 K.

Acknowledgements

This work was supported by Fundação para a Ciência e a Tecnologia (FCT), through IDMEC, under LAETA, project UID/EMS/50022/2013. Authors would like to thank Dr. Olivier Mathieu and Prof. Peter Glarborg for sharing the original files of their kinetic mechanisms.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2019.02.102>.

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